Unit: Unit 8: Quadratics **Name:** ____________________

Topic: Solving by Factoring and Other Methods **Date:** ____________

Advanced (Habanero)

Q1. Solve the equation $x^2 + 5x + 6 = 0$ by factoring.
(A) $x = -2, x = -3$
(B) $x = -1, x = -6$
(C) $x = 2, x = 3$
(D) $x = 1, x = 6$
(E) $x = -2, x = 3$

Q2. Solve $x^2 - 7x + 12 = 0$.
(A) $x = 3, x = 4$
(B) $x = 5, x = 6$
(C) $x = 3, x = 12$
(D) $x = 4, x = 7$
(E) $x = 5, x = 12$

Q3. Solve $x^2 + 2x - 8 = 0$.
(A) $x = 2, x = -4$
(B) $x = -1, x = 8$
(C) $x = 4, x = -2$
(D) $x = -4, x = 2$
(E) $x = -2, x = -8$

Q4. Verify if $x = -3$ is a solution to $x^2 + 4x + 3 = 0$.
(A) Yes
(B) No
(C) Maybe
(D) None
(E) Not enough info

Q5. Solve $2x^2 - 5x - 3 = 0$.
(A) $x = 3, x = -1$
(B) $x = -3, x = 1$
(C) $x = 2, x = -3$
(D) $x = -2, x = 3$
(E) $x = 3, x = -3$

Q6. Solve $x^2 - 9x + 20 = 0$.
(A) $x = 4, x = 5$
(B) $x = 5, x = 9$
(C) $x = 4, x = 9$
(D) $x = 4, x = 20$
(E) $x = 5, x = 20$

Q7. Solve $3x^2 + 2x - 1 = 0$.
(A) $x = -1, x =$
rac13
(A) $x = 1, x = -$
rac13
(A) $x =$
rac13, x = 3
(A) $x = -1, x = -3$
(B) $x = 1, x = 3$

Q8. Solve $x^2 + 10x + 24 = 0$.
(A) $x = -4, x = -6$
(B) $x = -6, x = -4$
(C) $x = 4, x = 6$
(D) $x = 6, x = -4$
(E) $x = -6, x = 4$

Open Ended Questions (FRQ)

Q9. Solve $2x^2 - 5x - 2 = 0$ and verify your solutions.

Q10. Solve $x^2 + 4x + 4 = 0$ by factoring and verify your solution.

Chef's Tips

- Tip 1: Make sure students show all work and explain their reasoning.
- Tip 2: Encourage students to check their solutions by plugging them back into the original equation.
- Tip 3: Remind students to simplify their answers and verify the solutions.